

Oscillation Charging Dynamics in Nanopore Supercapacitors with Organic Electrolyte

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Cite This: *ACS Appl. Mater. Interfaces* 2023, 15, 51274–51280



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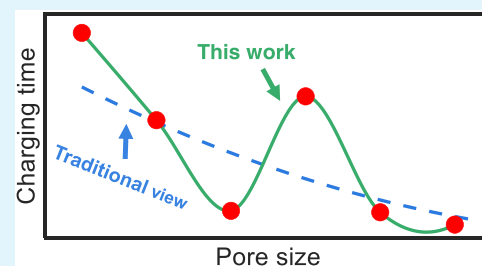
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Supporting Information

ABSTRACT: Nanopore electrodes have the potential to enhance the energy density of supercapacitors but tend to reduce charging dynamics, consequently impacting power density. A comprehensive understanding of their charging mechanisms can provide insights into how to boost charging dynamics. In this work, we conducted constant-potential-based molecular dynamics simulations to explore the charging mechanism of nanopore supercapacitors with organic electrolytes. Contrary to the traditional understanding associating larger pore sizes with faster charging, our results found a complex oscillatory behavior of the charging rate, correlating with nanopore size in organic electrolytes. An anomalously increased charging dynamics was found in the 0.9 nm pore. This anomalous enhancement can be attributed to the improved in-pore ion diffusion and reduced desolvation energy, owing to the orientation transition of the solvate molecules. These results pave a new way for innovative designs of nanoporous electrode supercapacitors that can enlarge both power and energy densities.

KEYWORDS: supercapacitors, porous electrodes, organic electrolytes, charging dynamics, desolvation, ion transport



INTRODUCTION

Supercapacitors have attracted significant attention due to their high power density, fast charging dynamics, and outstanding cycling life.^{1–3} Nevertheless, the energy density of supercapacitors is much lower than that of Li-ion batteries.^{2–4} Nanoporous electrodes could improve the energy density of supercapacitors owing to their large specific surface area and high electrical conductivity,^{5–7} perceived as the most encouraging material for supercapacitors. In particular, Gogotsi and co-workers found that the specific capacitance of supercapacitors exhibits an anomalous increase in subnanometer pores.^{8–10} This has garnered widespread interest in nanoporous supercapacitors.^{11–13} However, the nanoconfinement from porous electrodes would sluggish the ion transport, thus diminishing the power density of the supercapacitor.^{14–16} For example, Pean et al. revealed that the ion diffusion inside a nanoporous is slower by 1 order of magnitude than in neat liquids.¹⁷ An *in situ* NMR measure found that the diffusion of ions in nanoporous is reduced by over 2 orders of magnitude compared with bulk electrolyte.¹⁸ More importantly, a large number of experimental works^{19–22} revealed that a smaller pore results in slower charging. Therefore, subnanometer pores will reduce the charging dynamics and power density of supercapacitors.

Such experimental observations have prompted numerous modeling efforts and molecular simulations to investigate the charging mechanism in subnanometer pores.^{23–25} For example, Breitsprecher et al.^{26–28} examined the charging mechanism of supercapacitors with ionic liquids and nano-

pores through a combination of experimental measurements and molecular dynamics simulations. Notably, their findings revealed that ion transport and charging can be boosted by a slower scan rate. Siqueira and co-workers^{29–31} studied the charging dynamics of supercapacitors in different ionic liquids by constant-potential-based molecular dynamics simulation. Their results establish the correlation between charging dynamics and the in-pore ion structure. Surprisingly, our previous simulations present a trend different from existing findings, showing that the charging speed of supercapacitors with silt-like nanopores and ionic liquids oscillates with the pore size and, markedly, accelerates at specific pore sizes.¹⁶ Our simulation and experiments present contrasting trends, potentially stemming from variations in the electrode characteristics between the two approaches. The electrodes in experiment feature a complex pore size distribution, which could mask the nonmonotonic trends observed in simulations. Yet, recent molecular dynamics simulations have reaffirmed the conventional view by demonstrating that larger nanopores facilitate faster charging in supercapacitors with silt-like nanopores and aqueous electrolytes,³² suggesting that the electrolyte has a significant impact on that charging trend.

Received: August 29, 2023

Revised: October 7, 2023

Accepted: October 10, 2023

Published: October 25, 2023



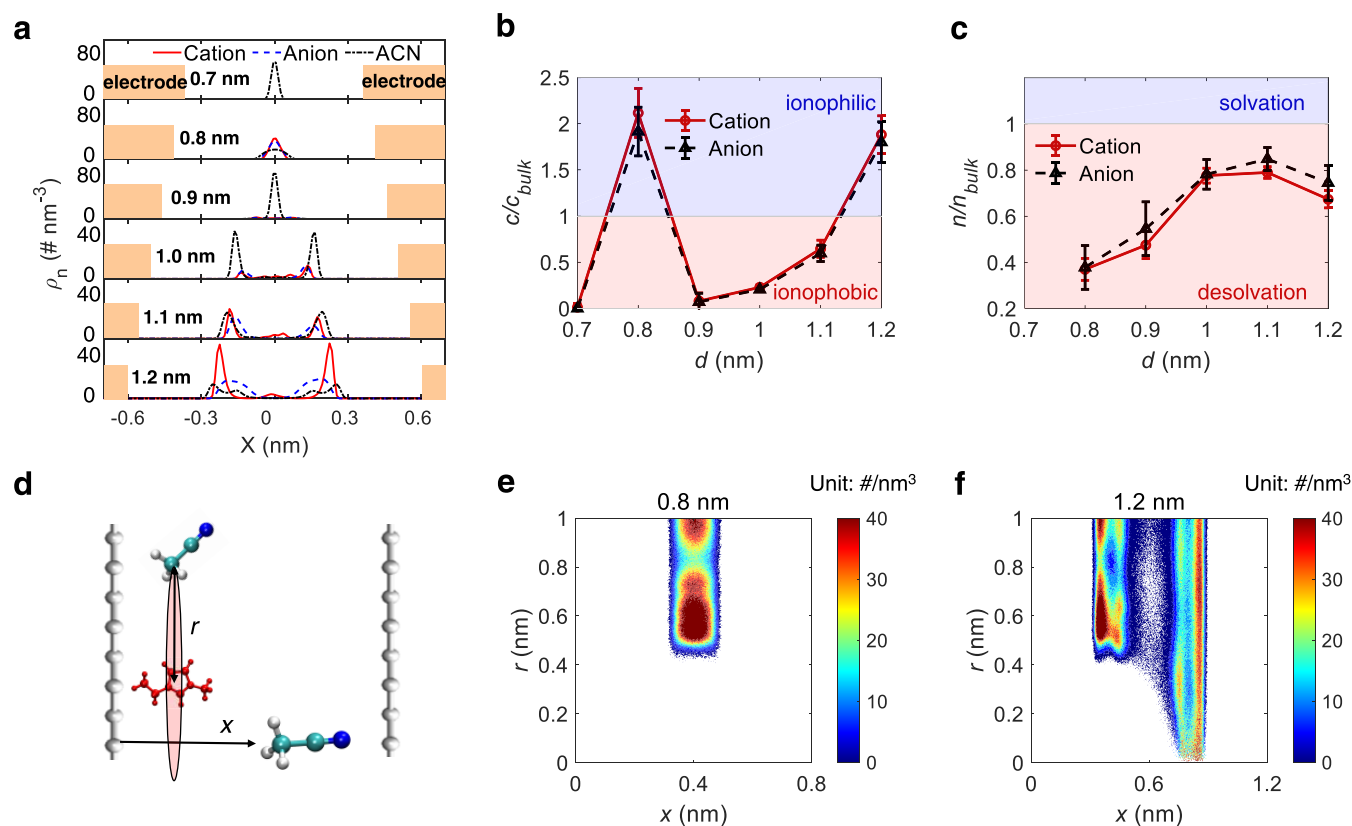


Figure 1. Ion structure inside nanopore at PZC. (a) Number density, ρ_n , of ions across slit pores with various pore sizes. The number of ACN is scaled by 1/15 for a better illustration. (b) In-pore ion mole fraction as a function of pore size. The mole fraction is expressed in terms of that of bulk (C_{bulk}). (c) In-pore ion solvation number as a function of the pore size. The solvation number is expressed in terms of the bulk (n_{bulk}). The solvation numbers of bulk cations and anions are 17.2 and 8.63, respectively. (d) Schematics of distributions of ACN molecules around a cation. r is the radial distance from the ion, and x is the distance along the pore width. (e, f) Pseudo-3D number density distributions of ACN molecules around a cation in 0.8 nm (e) and 1.2 nm (f) pore. $r = 0$ and $x = 0.4$ are the location of the cation.

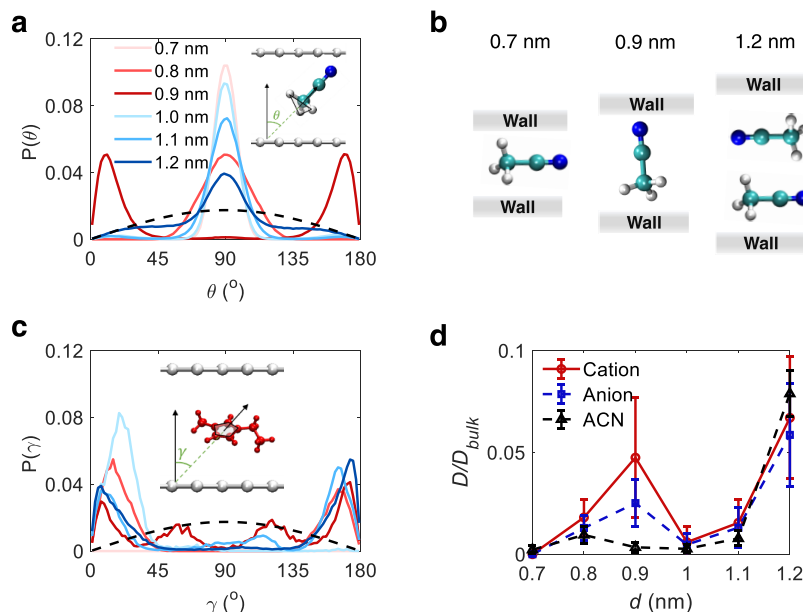


Figure 2. Ion orientation and diffusion inside the nanopore at PZC. (a) Angle distribution, $P(\theta)$, between the plane of an ACN and the pore wall. The inset is a schematic of the ACN orientation. (b) Schematics of the ACN orientation in different pores. (c) Angle distribution, $P(\gamma)$, between the plane of a cation and the pore wall. The inset is a schematic of the cation orientation. The black dashed line is the data from the bulk liquid. (d) Ion self-diffusion coefficient as a function of pore size. Ion diffusion coefficient is expressed in terms of that at the bulk (D_{bulk}). The diffusion coefficients of bulk cations, anions, and ACN are 1.28×10^{-9} , 1.18×10^{-9} , and $2.39 \times 10^{-9} \text{ m}^2/\text{s}$, respectively.

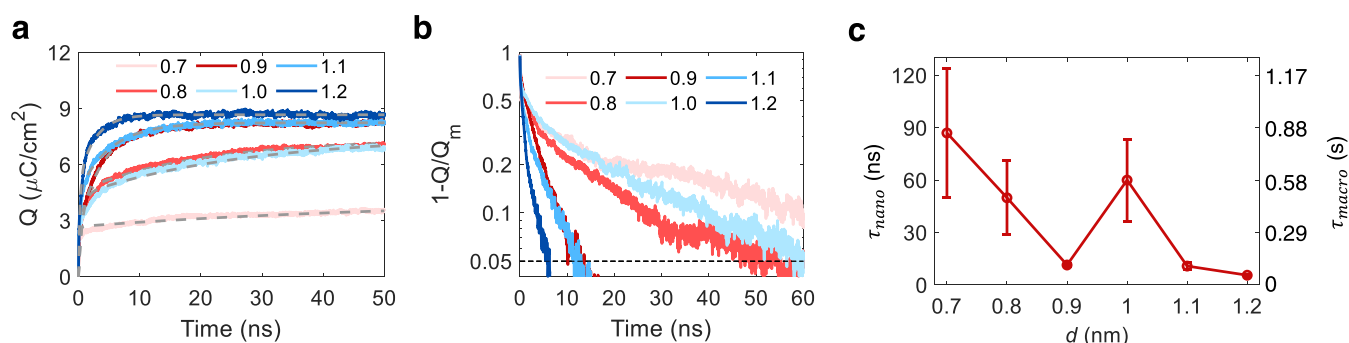


Figure 3. Charging dynamics. (a) Charge density of the pore throughout time. The data obtained from the simulation (solid lines) are compared to the equivalent circuit-model-fit curves (gray dashed lines). (b) Semilog plot of $1 - Q/Q_m$ during the charging process. The dashed line represents the time taken to reach 95% of the charging, i.e., τ_{nano} . (c) Charging time on a nanoscale (left y-axis) and macroscale (right y-axis) vs pore size.

However, the relationship and mechanism between the charging speed of nanopore supercapacitors and the electrolyte type remain ambiguous.

Here, we perform all-atom molecular dynamics simulations of 1.0 M 1-ethyl-3-methylimidazolium tetrafluoroborate ([EMIM][BF₄]) in acetonitrile (ACN) to study the charging dynamics of organic solution in different pore sizes (Figure S1 and Methods section). Different from the results of previous experiments,^{19–22} our findings uncover a novel result in solution electrolyte: the charging time exhibits a complex oscillatory with respect to nanopore size. Our simulation further demonstrates that the complex charging dynamics can be ascribed to the electrolyte structural transition caused by the pore size and electrode polarization.

RESULTS AND DISCUSSION

Ion Structure Inside Nanopores. We begin our analysis by inspecting the microstructure of electrolytes inside nanopores at a potential of zero charge (PZC). As illustrated in Figures 1a and S2, the variation in pore size leads to a transition in the structures of solvent molecules and ions within the nanopores, shifting from a single-layer to a multilayer arrangement: there is only a single layer of electrolytes accumulating at the central plane of the pores in 0.7 and 0.8 nm pores, whereas it transits into a two-layer distribution in pores with sizes larger than 1.0 nm. It is worth noting that the ion concentration within the pore exhibits oscillations with variations in pore size (Figure 1b), displaying a similar trend to previous theory and simulation results.³³ Especially, in the 0.8 and 1.2 nm pores, the ion concentration is higher than that in the bulk electrolyte, that is, the ionophilic pore. Conversely, in other pores, the ion concentration is lower than that in the bulk electrolyte, defining as ionophobic pores. The 0.7 and 0.9 nm pores are nearly unwetted by ions, demonstrating strong ionophobic or solvophobic.

However, regardless of whether it is an ionophilic or ionophobic pore, the number of ions solvated within the pore is reduced compared with the bulk electrolyte. This implies that partial desolvation occurs when ions are adsorbed into the nanopores. The quasi-three-dimensional ion solvation structure allows for a clear observation of the partial desolvation phenomenon (Figure 1d). At smaller pore sizes (0.8 nm), the cation surrounding ACN is mainly distributed along the length of the pore; the other parts are desolvated (Figure 1e). For the larger 1.2 nm pore, ACN molecules can distribute in more areas, showing a semispherical solvation shell (Figure 1f). This

also explains why smaller pore sizes display more intense desolvation (Figure 1c).

The microstructure inside the pore could be further understood by their orientations. Owing to the nanoconfinement, the angles of ACN molecules are close to 90° in most pores (Figure 2a,2b). While, in 0.9 nm pore, the orientation of ACN molecules is close to 0 and 180°, meaning that the carbon chain of ACN molecules is perpendicular to the wall (Figure 2a,2b). Those perpendicular orientations make ACN packing tighter and ordering more efficient (Figures 1a and S5a), reducing the diffusion of in-pore ACN molecules (Figure 2d). However, the cations within the 0.9 nm pore closely resemble a random distribution compared with the other pores (Figure 2c), thus contributing to faster ion diffusion (Figure 2d).

Effect of Pore Size on Charging Dynamics. We then focus on the charging dynamics, characterized by the evolution of the net charge, $Q(t)$, accumulated on nanopores. As shown in Figure 3a, the surface charge of the electrode exponentially increases with time. Figure 3b further illustrates that the charging process can be broadly divided into two stages. The first stage, driven by in-pore diffusivity, enables charging to 80% in a short time. The second exponential stage is attributed to ion adsorption and desorption, which are much slower, requiring a longer time to reach equilibrium. These observations align with previous charging results on ionic liquids¹⁴ and aqueous solutions.²²

The charging processes of nanopores can be described by the equivalent circuit model (Figure S6)¹⁵

$$Q(t) = Q_{\infty}(1 - A_1 e^{-t/\tau_1} - A_2 e^{-t/\tau_2}) \quad (1)$$

where Q_{∞} is the surface charge on the electrode when it is completely charged, and A_1 , A_2 , τ_1 , and τ_2 are fitting parameters (details in Supporting Information Part 3). It can be seen that the charging curves obtained from MD simulations fit extremely well with eq 1 (Figure 3a). The charging time, τ_{nano} , is evaluated by the time when the electrode charge reaches 95% Q_{∞} (dashed line in Figure 3b).

As illustrated in Figure 3c, in general, larger pore sizes lead to faster charging. When the pore size increases from 0.7 to 1.2 nm, the charging time decreases from 87 to 6 ns. Surprisingly, the charging speed of the 0.9 nm pore anomalously enhances to 11 ns, outperforming the 50 ns for the 0.8 nm and 60 ns for the 1.0 nm. Therefore, the charging time of supercapacitors with organic electrolytes exhibits oscillatory changes with the nanopore size. This differs from the traditional view that larger pore sizes result in faster charging^{14,15,19–21,34} but aligns with

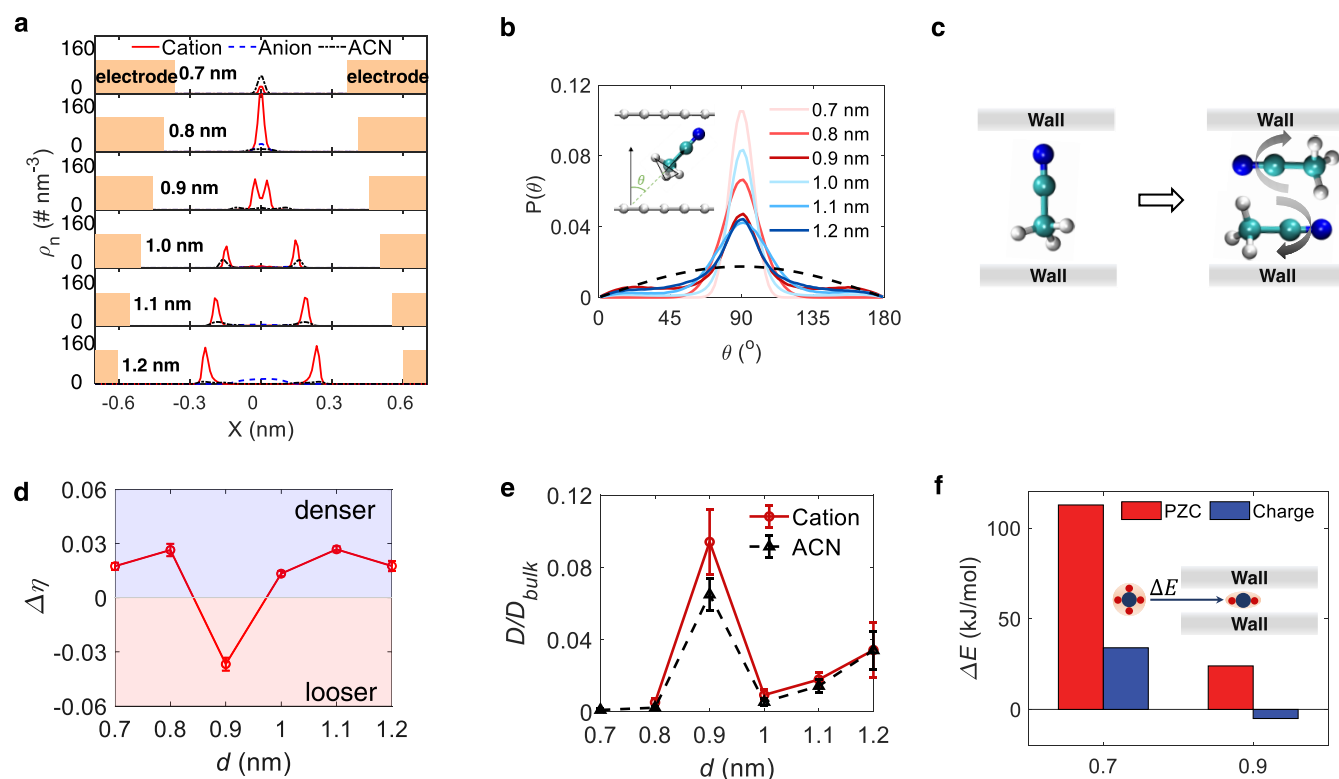


Figure 4. Ion structure after being charged. (a) Number density, ρ_n , of ions across slit pores with various pore sizes under negative polarization. The number of ACN molecules is scaled by 1/15 for a better illustration. (b) Angle distribution, $P(\theta)$, between the plane of an ACN and the pore wall under negative polarization. The inset is a schematic of the ACN orientation. The black dashed line is the data from bulk liquid. (c) Schematics of ACN orientation transition inside the 0.9 nm pore. (d) Packing factor inside the pore. The packing, η , is calculated by $\frac{V_{\text{electrolyte}}}{V_{\text{pore}}}$, where $V_{\text{electrolyte}}$ is the volume of electrolyte inside pore and V_{pore} is the volume of pore. $\Delta\eta$ represents the difference between the packing after being charged and at PZC. (e) Ion diffusion inside the pore. The diffusion coefficient is expressed in terms of the diffusion coefficient in the bulk (D_{bulk}). (f) Energy barriers for ion adsorption into pores.

our earlier findings in ionic liquids.¹⁶ While the charging time of organic electrolytes is much faster than that of pure ionic liquid (Figure S7). Our results further demonstrate that faster-charging 0.9 and 1.2 nm pores can deliver both higher energy and power densities (Figures S8–S9).

Indeed, the charging time of the nanopore is related to its length. The above charging time, τ_{nano} , with the order of magnitude of about 10 ns was obtained via MD simulations, in which the length of the electrode (l) is 6.4 nm. The charging time for a practical electrode (τ_{macro}) can be estimated by $\tau_{\text{macro}} = \tau_{\text{nano}} \times \left(\frac{l_{\text{macro}}}{l}\right)^2$,³⁵ where l_{macro} is the length of the electrode in macroscale. For electrodes with a thickness of 20 μm , τ_{macro} with pore sizes of 0.7 nm will be 0.95 s, which agrees with the experimental data with the order of magnitude 1 s.^{21,22}

Accelerated Ion Transport. Both 0.7 and 0.9 nm pores are characterized as ionophobic (Figure 1b). However, the charging time of the 0.7 nm pore is much slower than that of the 0.9 nm pore (Figure 3c). As shown in Figures 4a and S10, the charging processes of those pores are controlled by counterions adsorption, exhibiting a similar property to ionophobic pores.^{14,16} A larger number of counterions were rapidly adsorbed into the 0.9 nm pore (Figure S10), resulting in fast charging dynamics and high energy storage (Figure 3a). This is consistent with previous conclusions showing that the ionophobic pores could obtain both high power and energy densities,^{14,16,36} whereas the 0.7 nm pore exhibits an opposite

trend with a slow and limited ion adsorption, resulting in subpar charging dynamics and energy storage.

In the 0.7 nm pore, the ACN molecules exhibit a more pronounced preference for a 90° distribution, which deviates markedly from the distribution in the bulk region (Figure 4b). Meanwhile, the ACN molecules show a strong ordering distribution (Figure S4b). As a result, ion transport is substantially slowed (Figure 4e). While, within the 0.9 nm pore, the ACN distribution closely resembles that of the bulk region, indicating a smaller influence of nanoconfinement (Figure 4b). After being charged, the orientation of ACN molecules within the 0.9 nm pore transitions from an upright position to a lying-down configuration (Figures 4c and S5). This transition in orientation decreases the amount of solvent molecules that can be accommodated within the pore, leading to the expulsion of a greater quantity of ACN molecules during the charging process (Figure S10). During this process, ACN molecules also transit from an ordered distribution to a disordered distribution (Figure S5). The reduced packing of in-pore electrolyte (Figures 4d and S11) and disordered distribution (Figure S5b) greatly accelerate the ion diffusion (Figure 4e), thus speeding up the first stage of the charging process, a phase primarily dictated by ion diffusion. Additionally, a consistent trend of faster ion diffusion with increasing pore size is observed in other pores (Figure 4e), explaining the overall conclusion that larger pore sizes result in faster charging (Figure 3c).

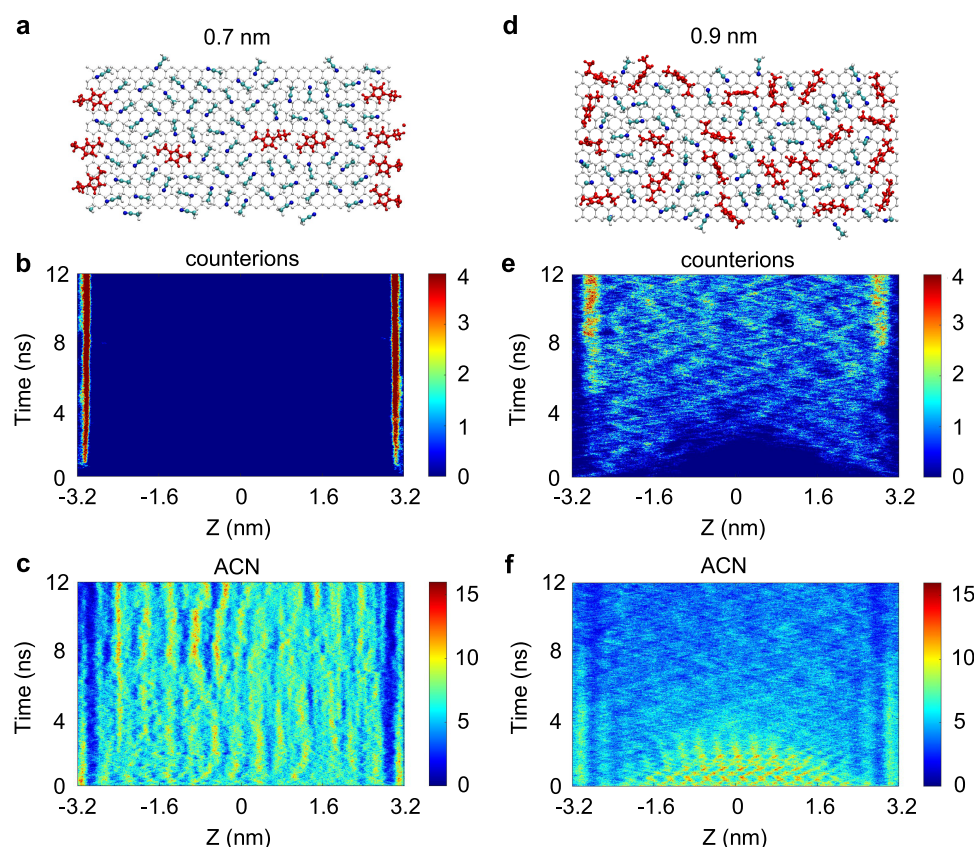


Figure 5. Ion motion path. (a) Top-view snapshot of 0.7 nm pore. (b, c) Time evolution of cation (b) and ACN (c) inside 0.7 nm pore along the direction of the pore length, Z. (d) Top-view snapshot of 0.9 nm pore. (e, f) Time evolution of cation (e) and ACN (f) inside 0.9 nm pore along the direction of the pore length, Z. Gray spheres are carbon atoms. Cation, anion, and acetonitrile are shown in red, blue, and white, respectively. Z = 0 is the center of the pore along the direction of the pore length. Unit of color bar: $\text{\#}/\text{nm}^3$.

Owing to the partial desolvation (Figure 1c), ions are required to overcome energy barriers during their adsorption into nanopores (Figure 4f, details are in Supporting Information Part 4). At PZC, large energy barriers are detected (~ 113 kJ/mol for 0.7 nm and 24 kJ/mol for 0.9 nm). Therefore, it is a challenge for ions to be adsorbed into pores, which elucidates ionophobic properties for those pores. After being charged, the energy barrier for the 0.9 nm pore decreases to -4.8 kJ/mol, while that of the 0.7 nm pore remains notably high at 35 kJ/mol.

Figure 5 illustrates the changes in the number density of counterions and solvent molecules during the charging process. A significant quantity of ACN is present within the 0.7 nm pore, with only a scant number of ACN molecules near the pore entrance desorbed after being charged (Figure 5a,c). This finding clarifies the reason for the substantial energy barrier observed for ion adsorption (Figure 4f). Consequently, a limited number of counterions are adsorbed into the pore entrance, while the adsorption of ions into the pore center can be negligible (Figure 5a,b). This constraint greatly deteriorates the charging process.

In contrast, in the 0.9 nm pore, a substantial amount of ACN molecules is expelled from the pore, starting from the pore entrance and extending toward the pore center (Figure 5f). Subsequently, a large number of counterions are rapidly adsorbed into the pore entrance from the bulk region and diffuse toward the center of the pore (Figure 5d). The acceleration in ion adsorption enhances the second stage of charging.

CONCLUSIONS

We investigated the charging mechanism of nanopore supercapacitors with organic electrolytes by constant-potential-based molecular dynamics simulations. Our results revealed a new phenomenon in organic electrolytes, showing that the charging rate oscillates with nanopore sizes. The charging dynamics is anomalously increased in the 0.9 nm pore. This anomalous enhancement can be attributed to the structural transition of the in-pore solvent molecules after being charged. The structural transition not only reduces the in-pore packing of electrolytes, thus accelerating ion diffusion, but also lowers the energy barrier for ions adsorbed.

Our previous research revealed that in ionic liquid supercapacitors, the charging rate displays oscillatory behavior in relation to nanopore size.¹⁶ In this work, we provide further confirmation of these oscillatory charging dynamics in relation to the nanopore size within organic electrolytes. The oscillatory charging is due to the structure translation of the in-pore electrolyte, induced by nanoconfinement and polarization. However, in aqueous electrolytes, the charging behavior follows with the conventional understanding: larger pore sizes result in faster charging.³² That could be ascribed to that in-pore ion concentration in electrolytes does not oscillate in correlation with pore sizes,³² suggesting that the aqueous electrolyte structural translation induced by pore size is less evident in comparison with organic electrolytes.

The oscillatory behavior of charging has yet to be reported through an experimental measurement. This could be

attributed to the intricate nature of the pore size distribution in experimental porous electrodes, causing the phenomenon of charging rate oscillation with pore size to be obscured. To empirically substantiate this phenomenon, it would be advantageous to utilize porous electrodes with a singular, adjustable pore size distribution, such as conductive metal–organic frameworks (c-MOFs)³⁷ and covalent organic frameworks (COFs).³⁸

METHODS

As shown in Figure S1a, each MD simulation system has two nanopores as the electrodes, separated by two IL reservoirs. The size of the pore (d) is defined as the center-to-center distance between walls, ranging from 0.7 to 1.2 nm. The nanopore and bulk liquid lengths are 6.4 and 9.6 nm, respectively. The all-atom force fields of [EMIM][BF₄] and ACN were employed (Figure S1b). Especially, the force field of [EMIM][BF₄] was taken from Doherty et al.,³⁹ and the force field of ACN was taken from Wu et al.⁴⁰ The Lennard-Jones model of carbon electrode atoms⁴¹ was used. All simulations were carried out by a modified MD code based on the GROMACS package.⁴² The electrolyte temperature in the NVT ensemble was controlled at 298 K by using the v-rescale thermostat.

The applied potential between two nanopore electrodes in our simulations is employed by the constant-potential method (CPM), accommodating for the fluctuations of charges in electrode atoms throughout simulations.^{13,43,44} The system was first kept for 50 ns for equilibration without any applied potential. Subsequently, the potential difference between the two electrodes was set at 3.0 V, and the system could equilibrate enough to guarantee that the system ultimately reached equilibrium (>50 ns, Figure 3a). Each charging process was independently carried out more than 15 times to reduce statistical error.

The ion diffusion coefficient along the direction of the pore length was calculated using

$$D = \lim_{t \rightarrow \infty} \frac{1}{2nt} \langle |r_i(t) - r_i(0)|^2 \rangle \quad (2)$$

where $r_i(t)$ is the position of the ion i at time t , $r_i(0)$ is the position of the ion i at $t = 0$, and n is the dimensionality. Here, we only consider the direction of the pore length because the charging dynamics is mainly decided by the ion moving along the pore length.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.3c12913>.

Details of molecular dynamics simulation, ion transport, and ion structure inside the pore (PDF)

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Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors thank the Basic Ability Promotion Project for Yong Teachers in Guangxi No. 2023KY0019 and the Specific Research Project of Guangxi for Research Bases and Talents No. AD23026174. They also thank the high-performance computing platform of Guangxi University for providing HPC resources to accomplish simulations.

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